CLAIMS

1. A method of detecting a target analyte and/or performing a spatial multiplexed assay, the method comprising:

- providing a microplate comprising:

a plurality of wells arranged in a two-dimensional array, each of the wells comprising:

- a bottom surface,
- sidewalls extending from the bottom surface to form an open top; and
- at least two subwells in the bottom surface, the at least two subwells having sidewalls that extend below the bottom surface of the well, wherein each of the at least two subwells comprises a capture binding agent immobilized therein that is configured to bind to a target analyte, if present, in a sample;

- adding a sample to the well such that the sample fluidically contacts each of the at least two subwells, such that a target analyte, if present, binds to the capture binding agent in one or more of the at least two subwells, and wherein an upconverting nanoparticle (UCNP) labeled conjugate in contact with the sample binds with the target analyte, if present, such that detection of the labeled target analyte in a respective subwell may be used to determine if the target analyte is present in the sample;

- detecting a label of the upconverting nanoparticle (UCNP) labeled conjugate at one or more of the at least two subwells to thereby determine whether the target analyte is present in the sample; and

- determining a concentration of the target analyte based on a signal strength of a signal from the UCNP labeled conjugate, wherein the signal has a signal strength that corresponds to a concentration of the target analyte, if present, in the sample.

2. The method of Claim 1, wherein the capture binding agent comprises at least a first capture binding agent and a second capture binding agent in different ones of the at least two subwells.

3. The method of Claim 2, wherein the UCNP labeled conjugate comprises a first UCNP labeled conjugate and a second UCNP labeled conjugate that are configured to bind to

corresponding target analytes bound to the first and second capture binding agents, respectively, in the at least two subwells.

4. The method of Claim 3, wherein a UCNP of the first and second UCNP labeled conjugates is the same UCNP, and detecting a label of the labeled conjugate at one or more of the at least two subwells to thereby determine whether the target analyte is present in the sample comprises determining whether first and second target analytes are present based on which of the at least two subwells emit a signal from the UCNP labeled conjugate.

5. The method of Claim 4, wherein detecting a UNCP label of the UNCP labeled conjugate at one or more of the at least two subwells comprises impinging photons on the at least two subwells such that the UCNP labeled conjugate emits a signal.

6. The method of Claim 5, wherein at least one of the first and second capture binding agents comprises a control capture binding agent and other of the first and second capture binding agents comprises a test capture binding agent, wherein a signal strength of a UCNP labeled conjugate bound to the test capture binding agent relative to a signal strength of a UCNP labeled conjugate bound to the control capture binding agent corresponds to the concentration of the target analyte, if present, in the sample.

7. The method of Claim 1, wherein the capture binding agent comprises a first half of a binding pair and the capture binding agent is bound to a surface of the at least two subwells via binding of the first half of the binding pair to a second half of the binding pair on the surface of the at least two subwells.

8. The method of Claim 1, wherein the upconverting nanoparticle (UCNP) labeled conjugate is present on a bottom and/or side of the well prior to adding the sample to the well.

9. The method of Claim 1, further comprising adding the upconverting nanoparticle (UCNP) labeled conjugate to the sample.

10. The method of Claim 1, wherein the target analyte is human chorionic gonadotropin (hCG).

11. The method of Claim 1, wherein the target analyte is placental growth factor (PlGF).

12. A microplate comprising:

a plurality of wells arranged in a two-dimensional array, each of the wells comprising:

- a bottom surface;
- sidewalls extending from the bottom surface to form an open top; and
- at least two subwells in the bottom surface, the at least two subwells having sidewalls that extend below the bottom surface of the well;

wherein each subwell is sized and configured such that when, in use, a fluid sample can be added into the well and said sample fluidically contacts each of the at least two subwells,

and wherein each of the at least two subwells comprises a capture binding agent immobilized therein and that is configured to bind to a target analyte, if present, in the sample,

and wherein, in use, an upconverting nanoparticle (UCNP) labeled conjugate in the at least two subwells binds with the target analyte, such that detection of the labeled target analyte in a respective subwell may be used to determine if the target analyte is present in the sample.

13. The microplate of Claim 12, wherein the capture binding agent comprises at least a first capture binding agent and a second capture binding agent, respectively immobilized in different ones of the at least two subwells.

14. The microplate of Claim 13, wherein the UCNP labeled conjugate comprises a first UCNP labeled conjugate and a second UCNP labeled conjugate that are configured to bind to corresponding target analytes bound to the first and second capture binding agents, respectively, in the at least two subwells.

15. The microplate of Claim 14, wherein a UCNP of the first and second UCNP labeled conjugates is the same, and the microplate is configured to be analyzed by a reader that determines whether the first and second target analytes are present based on which of the at least two subwells emit a signal from the UCNP.